

Timothy James Wood

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SUMMARY

I am a first-class Biological Sciences graduate beginning an HDR UK-funded MSc in Bioinformatics at Bristol. I have completed two research projects in statistical population genetics under Prof. Mark Beaumont, covering demographic inference and genome-wide selection scans. I am seeking a PhD in statistical genetics and machine learning, with a particular interest in inference methodology for genomic data.

EDUCATION

University of Bristol

MSc Bioinformatics

Sept 2026 – Sept 2027

HDR UK MSc Scholarship (Coeliac UK)

- Dissertation (Prof. Mark Beaumont): simulation-based inference for the genetic architecture of coeliac disease

University of Bristol

BSc Biological Sciences

2023 – 2026

First Class Honours

- First-class marks in both the final-year research project and literature review

RESEARCH EXPERIENCE

Prof. Mark Beaumont's Group, University of Bristol

Jun – Sept 2026

Research Assistant, with Dr Dennis Prangle and Grace Yan

github.com/timj-wood/wildcat_models

- Built a folded joint site-frequency spectrum from 46 whole-genome Scottish wildcats and domestic cats
- Fitted three two-population models in dadi, including an 11-parameter domestication and land-bridge model
- Implemented composite-likelihood model selection and Godambe confidence intervals from first principles
- Showed that deep parameters lie on a single likelihood ridge
- Diagnosed a boundary optimum that invalidated the best-scoring model, leading to a defensible primary result
- Ran optimisation as SLURM array jobs on Bristol's HPC; delivered a technical report and repository
- Report and repository are intended as a baseline for later simulation-based inference work

BSc Research Project: Selection and Introgression in the European Wildcat

Sept – Dec 2025

Supervisor: Prof. Mark Beaumont

Data from Dr Jo Howard-McCombe, RZSS

- Processed whole-genome SNP data for 108 wild, captive and domestic cats (VCFtools, PLINK)
- Ran a pcadapt genome scan for selection; identified 1,275 outlier intervals and annotated 11 candidate genes
- Found significant GO enrichment for GABA, taurine and calcium-ion transport
- Developed PCA visualisations to separate introgression from demographic structure
- Critically evaluated drift vs. selection, linkage and threshold choices

BSc Literature Review: Engineering Predictability in Living Systems

Sept – Dec 2025

Supervisor: Prof. Thomas Gorochoowski

Biocompute Lab

- Reviewed synthetic circuits, genome recoding and biological foundation models (Evo, ESM-2, ProGen)
- Argued that data-driven models capture evolutionary correlation rather than causal mechanism
- Proposed context-aware data and adaptive feedback as remedies

TECHNICAL SKILLS

Programming Python (NumPy, SciPy, pandas), R (tidyverse), Bash, Git, SLURM/HPC, LaTeX

Statistics Composite likelihood, CLAIIC model selection, Godambe estimators, block bootstrap, PCA

Genomics Demographic inference (dadi), selection scans (pcadapt), variant QC (VCFtools, PLINK)

OTHER EXPERIENCE

PwC LLP

June – July 2025

Audit Intern

London

- Supported the audit of WPP plc (FTSE 100); produced analytical working papers from large financial datasets
- Trained in IFRS, audit methodology and proprietary data systems

University of Bristol Engineering Rugby Society

May 2025 – Apr 2026

President

- Ran club finances and Students' Union relations; secured sponsorship and event funding